

Module: Azure Blob Storage and Azure Files

Session Aim

To understand Azure Blob Storage and Azure Files, create and configure storage resources, and manage data using snapshots, versioning, soft delete, storage tiers, and lifecycle management.

1. Introduction

Azure Storage provides secure, scalable, and highly available cloud storage.

There are different storage services:

Storage Service	Purpose
Blob Storage	Store unstructured data
Azure Files	Managed file shares
Queue Storage	Message storage
Table Storage	NoSQL key-value storage

Today we focus on

- Azure Blob Storage
 - Azure Files
-

2. Azure Blob Storage

Definition

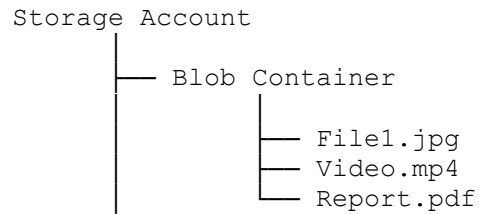
Azure Blob Storage is Microsoft's object storage service used for storing large amounts of unstructured data.

Examples

- Images
- Videos
- Documents
- Backup files
- Log files

- Virtual machine disks

Blob Hierarchy



Explain:

Storage Account → Container → Blob

Blob Types

Block Blob

Used for

- Images
- Documents
- Videos

Most commonly used.

Append Blob

Used for

- Logs
- Audit records

New data is appended at the end.

Page Blob

Used for

Virtual Machine disks (VHD)

Practical 1

Create a Blob Container

Steps

1. Login to Azure Portal
2. Open Storage Account
3. Select

Data Storage
→ Containers

4. Click

+ Container

5. Enter

Name:
studentfiles

6. Public Access

Private

7. Click Create

Students now have a blob container.

Upload a Blob

Open the container

Click

Upload

Select

sample.pdf

Click Upload

Students observe

`sample.pdf`

inside the container.

Download Blob

Select

`sample.pdf`

↓

Download

Delete Blob

Select Blob

↓

Delete

Explain that deletion can be recovered if Soft Delete is enabled.

3. Azure File Shares

Unlike Blob Storage,

Azure Files behaves like a normal Windows network drive.

It supports

- SMB protocol
 - NFS protocol
-

Real-world Example

Suppose five employees need to access the same folder.

Instead of

Laptop A

Laptop B

Laptop C

Create

Azure File Share

Everyone connects to the same shared folder.

Hierarchy

Storage Account



Practical 2

Create Azure File Share

Storage Account



File Shares



- File Share

Enter

Name

companyfiles

Choose quota

100 GB

Click Create

Upload files

↓

Browse

↓

Upload

Students see the file share working like a shared drive.

4. Storage Tiers

Azure provides different storage tiers to reduce cost.

Tier	Purpose
Hot	Frequently accessed data
Cool	Occasionally accessed
Cold	Rarely accessed (where available)
Archive	Long-term storage

Example

Company CCTV footage

Current month

↓

Hot

Last six months

↓

Cool

Older than one year



Archive

Demonstration

Storage Account



Container



Select Blob



Change Access Tier

Choose

- Hot
- Cool
- Archive

Students observe how storage cost changes depending on access frequency.

5. Blob Snapshots

A snapshot is a read-only copy of a blob at a specific point in time.

Example

Original File

Report.docx



Create Snapshot



Snapshot stored

If the original file changes,
the snapshot remains unchanged.

Uses

- Backup
 - Restore
 - Version comparison
-

Practical

Upload

`Resume.pdf`

Create Snapshot

Modify Resume

Upload again

Compare

- Current Blob
 - Snapshot
-

6. Soft Delete

Normally

`Delete`

↓

`Gone forever`

Soft Delete changes this.

`Delete`

↓

`Moved to recoverable state`

↓

Restore anytime within retention period

Configure Soft Delete

Storage Account

↓

Data Protection

↓

Enable

Blob Soft Delete

Choose

Retention

7 days

Delete a file

Students recover it from

Deleted Blobs

7. Blob Versioning

Versioning automatically creates a new version whenever a blob is modified.

Example

Report.docx

↓

Version 1

↓

Edit

↓

Version 2

↓

Edit

↓

Version 3

Old versions remain available.

Benefits

- Accidental overwrite protection
 - Rollback
 - Audit
-

Practical

Enable

Blob Versioning

Upload

Project.pdf

Replace with another version

Observe

Version History

Restore an older version.

8. Lifecycle Management

Lifecycle Management automatically moves or deletes blobs based on rules.

Instead of manually managing files,

Azure performs the task automatically.

Example

Rule

If file not used for

30 days

↓

Move to Cool Tier

90 days

↓

Move to Archive

365 days

↓

Delete

Benefits

- Saves storage costs
 - Automatic data management
 - Reduces manual work
-

Practical

Storage Account

↓

Lifecycle Management

↓

Add Rule

Example Rule

IF

Blob age > 30 days

THEN

Move to Cool Tier

Another Rule

Blob age >180 days

↓

Archive

Another Rule

Blob age >365 days

↓

Delete

Save the policy.

Explain that Azure evaluates these rules automatically.

Blob Storage vs Azure Files

Feature	Blob Storage	Azure Files
Storage Type	Object storage	File share
Protocol	REST API	SMB/NFS
Suitable For	Images, videos, backups, logs	Shared folders and application data
Folder Structure	Containers and virtual folders	Traditional file and folder hierarchy
Mount as Drive	No	Yes
Typical Users	Applications and cloud services	Users and applications needing shared file access

Complete Classroom Demonstration Flow

1. Create a Storage Account.
 2. Create a Blob Container.
 3. Upload, download, and delete blobs.
 4. Change the blob access tier (Hot/Cool/Archive).
 5. Enable Blob Soft Delete and recover a deleted blob.
 6. Enable Blob Versioning and upload multiple versions of a file.
 7. Create a Blob Snapshot and compare it with the current blob.
 8. Configure a Lifecycle Management rule to move older blobs to cooler tiers.
 9. Create an Azure File Share.
 10. Upload files to the file share and explain how it can be mounted as a shared network drive.
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Expected Learning Outcomes

By the end of this session, students should be able to:

- Explain the differences between Azure Blob Storage and Azure Files.
- Create and configure Blob containers and Azure File Shares.
- Upload, download, and organise data in Azure Storage.
- Select appropriate storage tiers (Hot, Cool, Cold, or Archive) based on access patterns.
- Configure and use blob snapshots, soft delete, and versioning for data protection.
- Create lifecycle management policies to automate storage optimisation and reduce costs.
- Identify suitable real-world scenarios for using Blob Storage versus Azure Files.